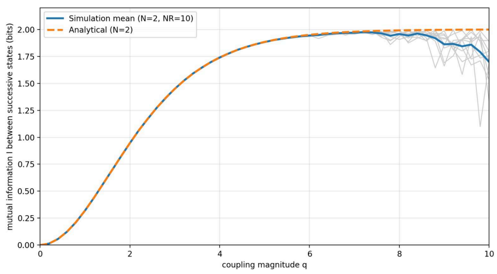
GOAL 1: Compare Analytical and Simulated MI in NRooks

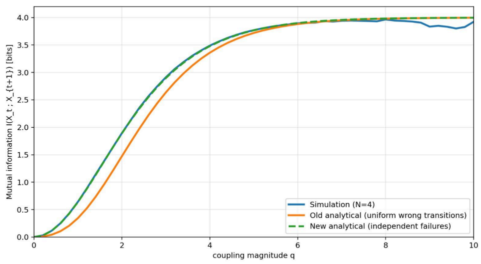
A plot of the mutual information I between successive states (0..10 bits) in a NRooks system, as a function of the magnitude q of the nonzero matrix elements (0..10).

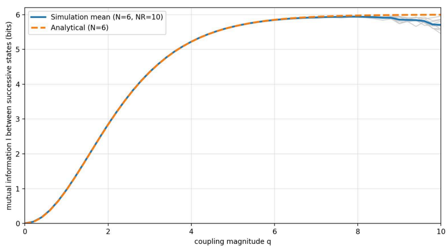
In the same plot, there should be shown the results for two system sizes: An RNN with only N=4 neurons (very good match expected) and another RNN with N=10 neurons (to see the limits of the numerics).

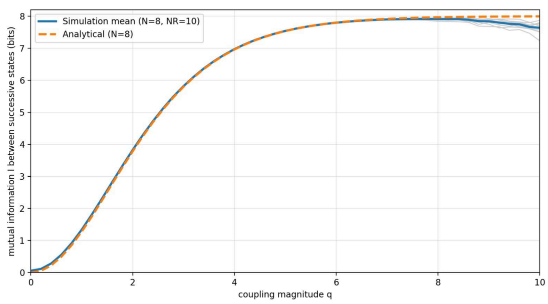
Each of the two I(q, fixed N) cases should be compared for fully analytical calculation (see NRooks Supplemental and associated program), for the transition matrix approach, and for the full dynamical simulation.

Use three different colors to distinguish the 'analytical', 'transition matrix' and 'simulation' methods.









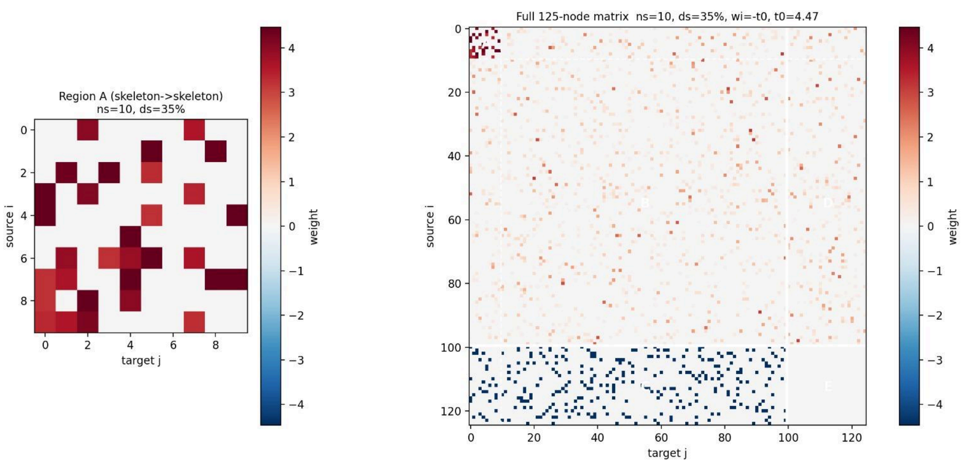
GOAL 2: Reproduce Prebeck system and MI curves

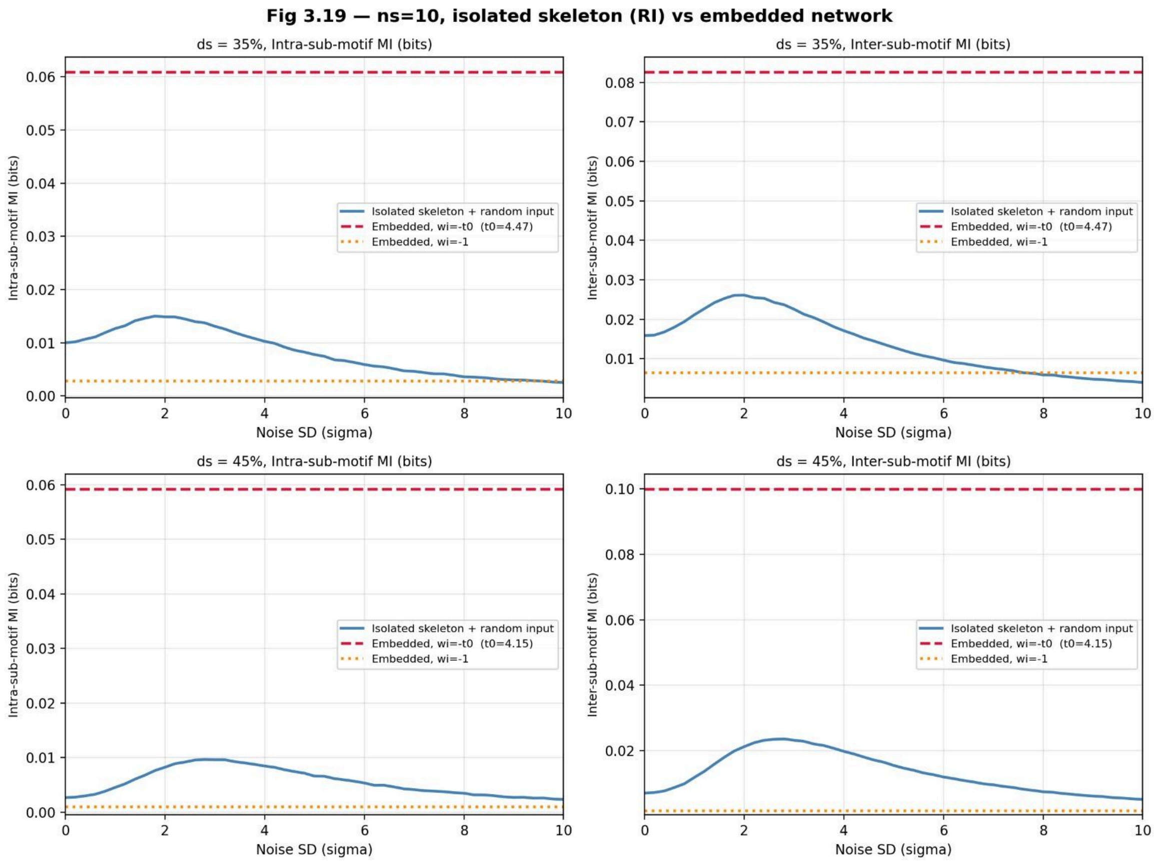
(1) A concise list of all the features of the weight matrix used for Fig. 3.19. in Karin Prebeck's thesis.

(2) A routine that creates such a weight matrix. The Python function should contain all the features from (1). We can later modify/simplify this routine.

(3) A plot of the weight matrix like Fig. 2.7 in Prebeck's thesis.

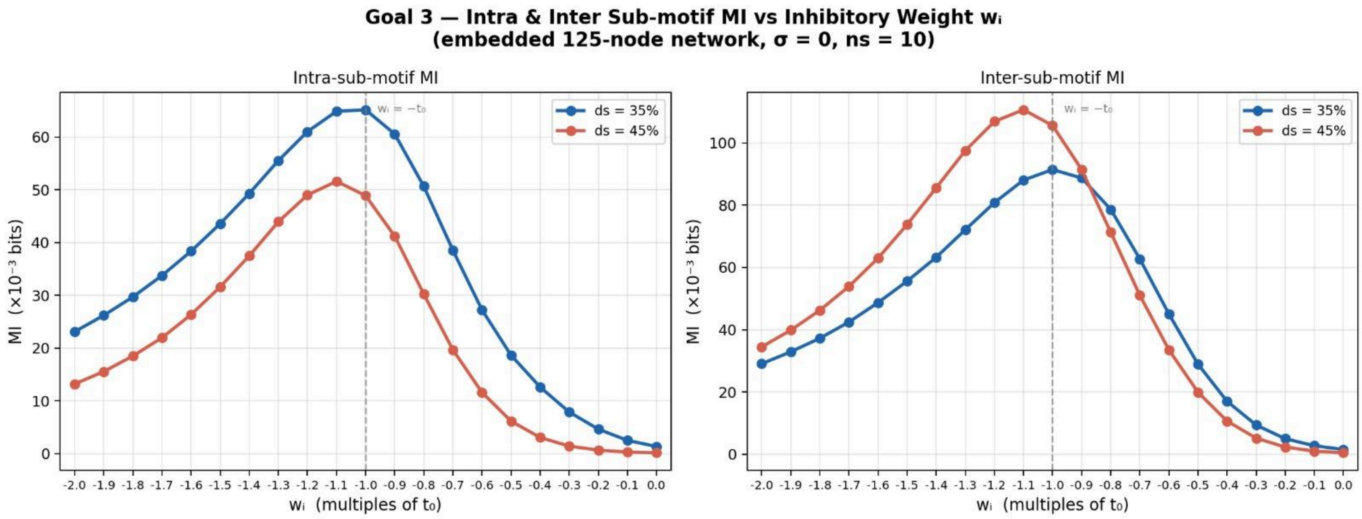
(4) A first plot of mutual information I for the isolated core as a function of added noise, and for the see-embedded core without added noise.





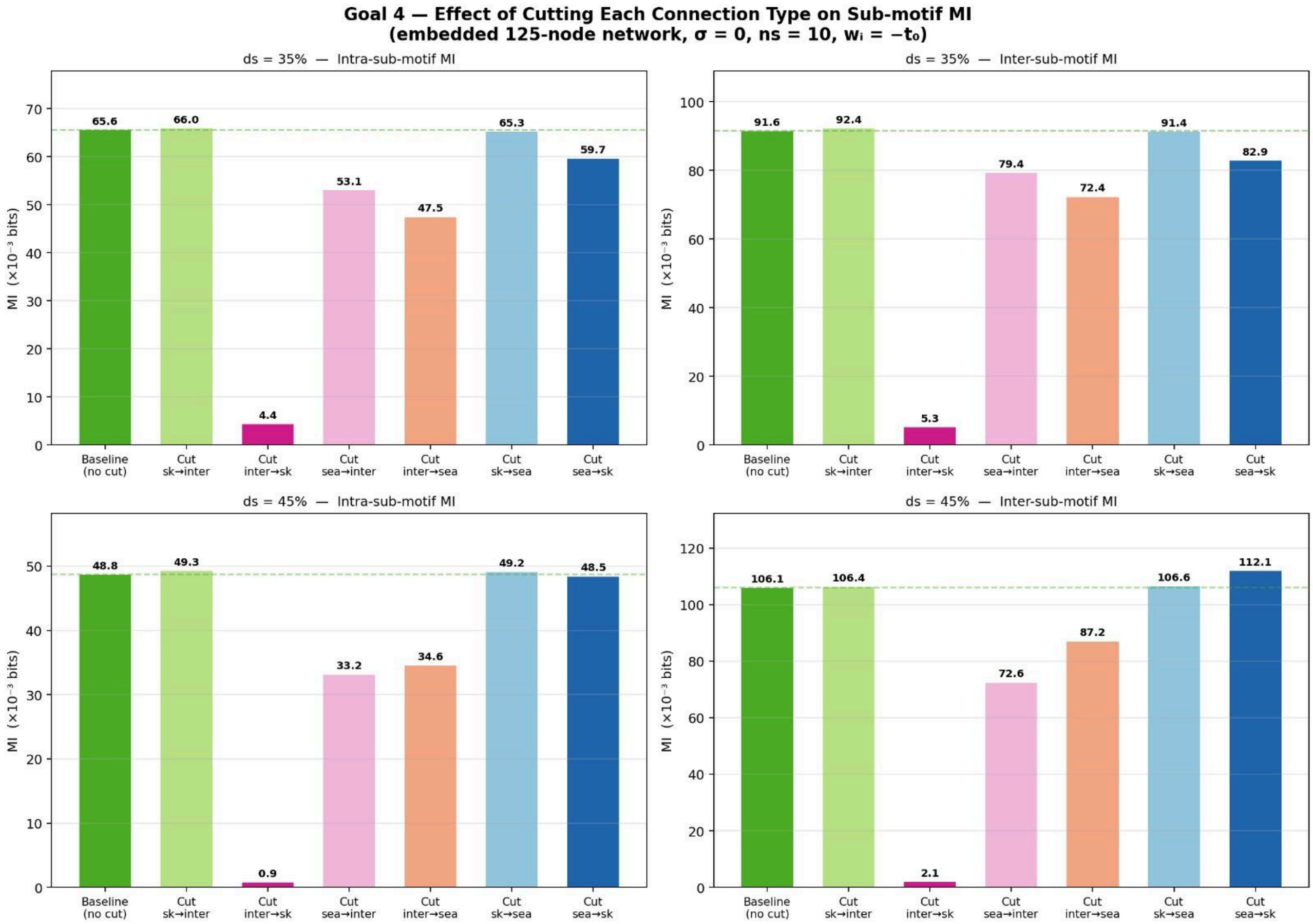
GOAL 3: MI as a function of inhibitrory weight

Ali, so far you have tested two values of the inhibitory weight wi. Could you please plot the MI of the embedded network as a funktion of wi, ranging from -2*t0 to 0 in at least 5 or more steps ? I would like to know if t0 is really the optimum.



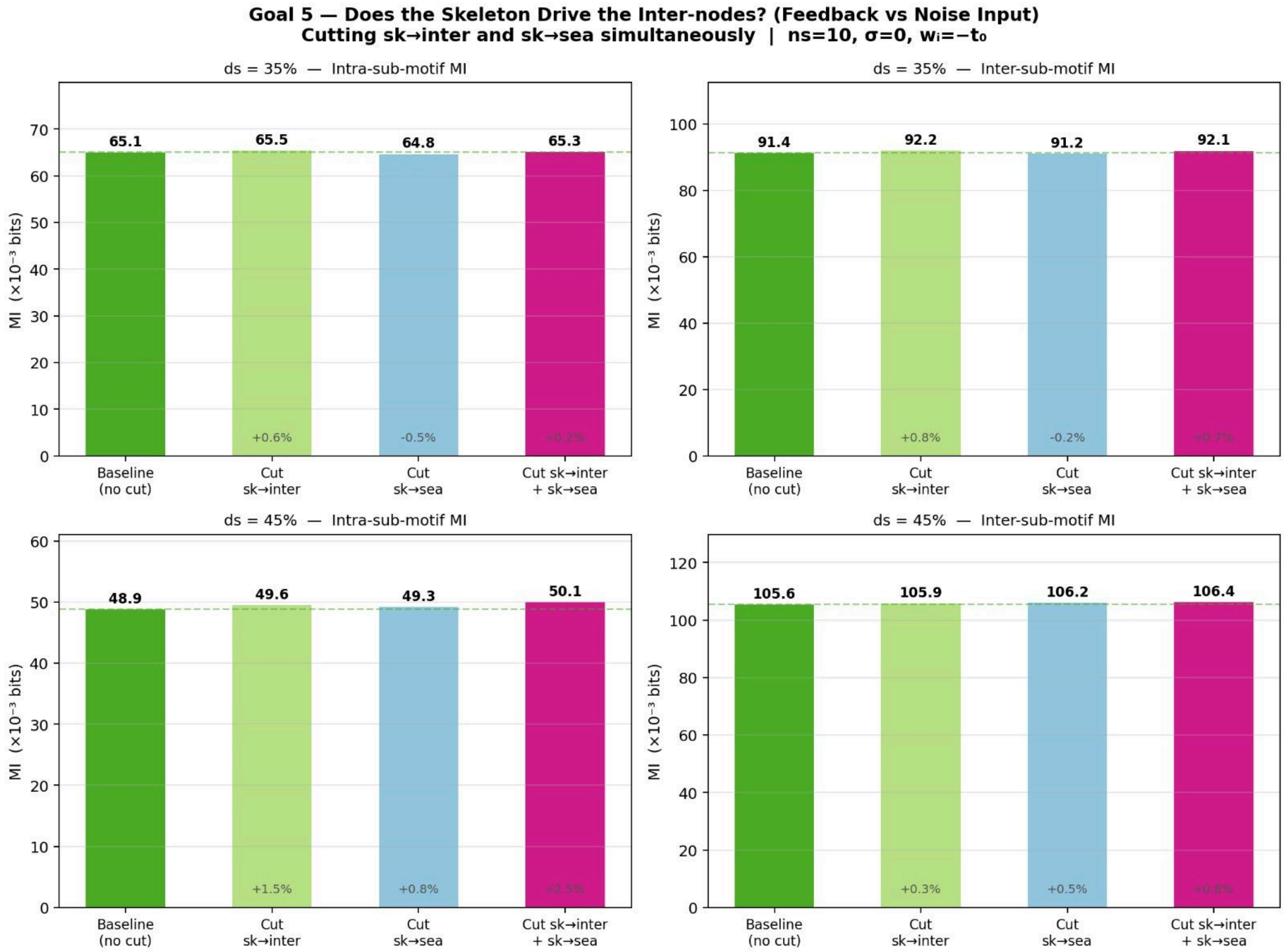
GOAL 4: Numerical Lesion Experiments

In this diagram each of the 7 arrows describes some specific type of connection. The strong self-connection of the skeleton is essential to make it an RNN. But the remaining 6 connection types could be systematically cut (set to zero). Could you compute the MI for all 6 possible cuts and present the resulting MIs as bar diagrams ?



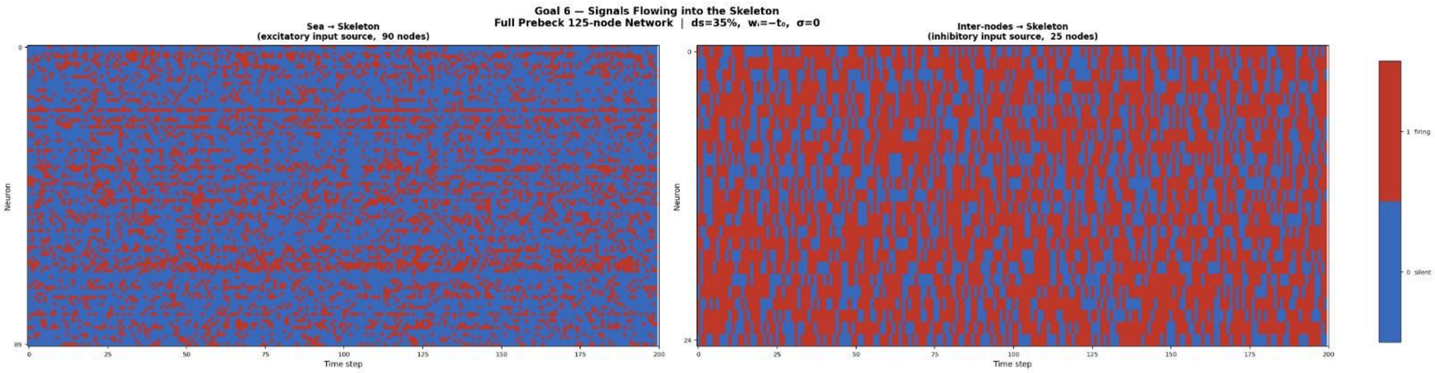
GOAL 5: Double Cut Experiments

Could you simultaneously cut sk->inter and sk->sea ? If this has no effect, it is clearly not a feedback loop but a sophisticated noise input. Am I correct in my argumentation ?



GOAL 6: Direct Vizualisation of Signals into Skeleton

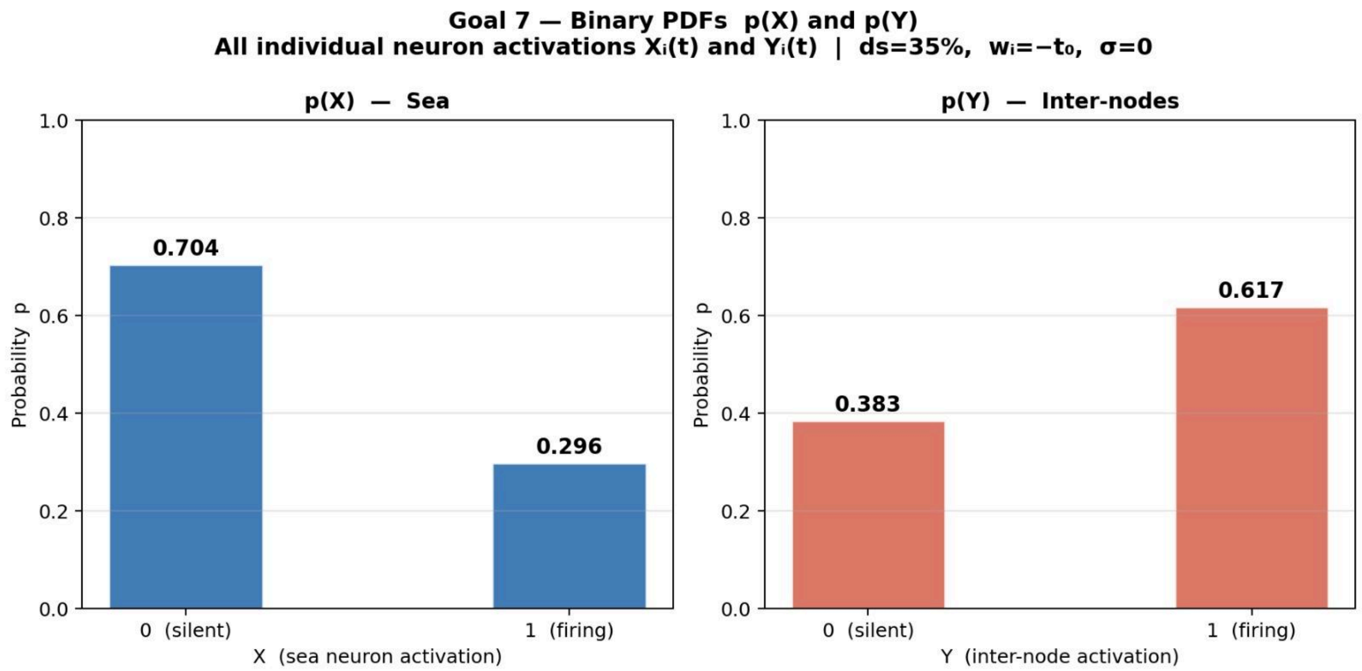
Bevor any PDF or Correlation analysis, I would like to see a plot of all the signals flowing into the skeleton from the rest of the system, in the form of a simple matrix plot like this.

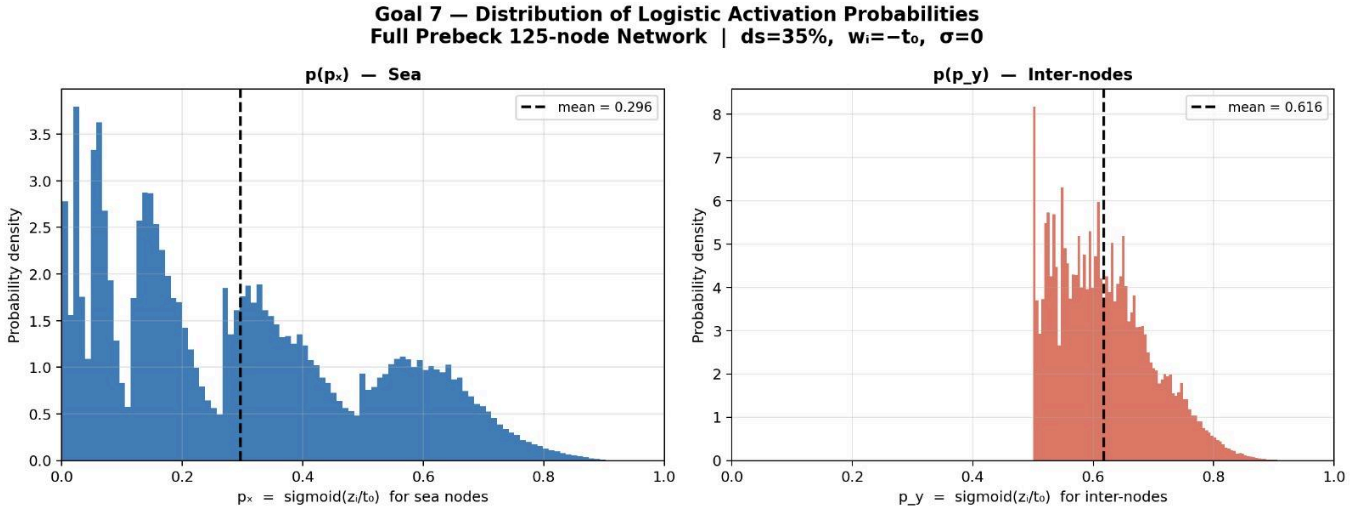


GOAL 7: Probability Density Functions of Activations in Sea and Inter

We can collect, for example, all the sea-activations X_i(t), for all sea-neurons i and all times t, in a single long list. From this list, we can compute the PDF/Histogram p(X), normalized to area one.

Please make two separate Plots p(X) and p(Y).





GOAL 8: Pearson Correaltion Matrices

Assume X[t,i] is the time series (t) of activations of neuron i in population X.

Assume Y[t,j] is the time series (t) of activations of neuron j in population Y.

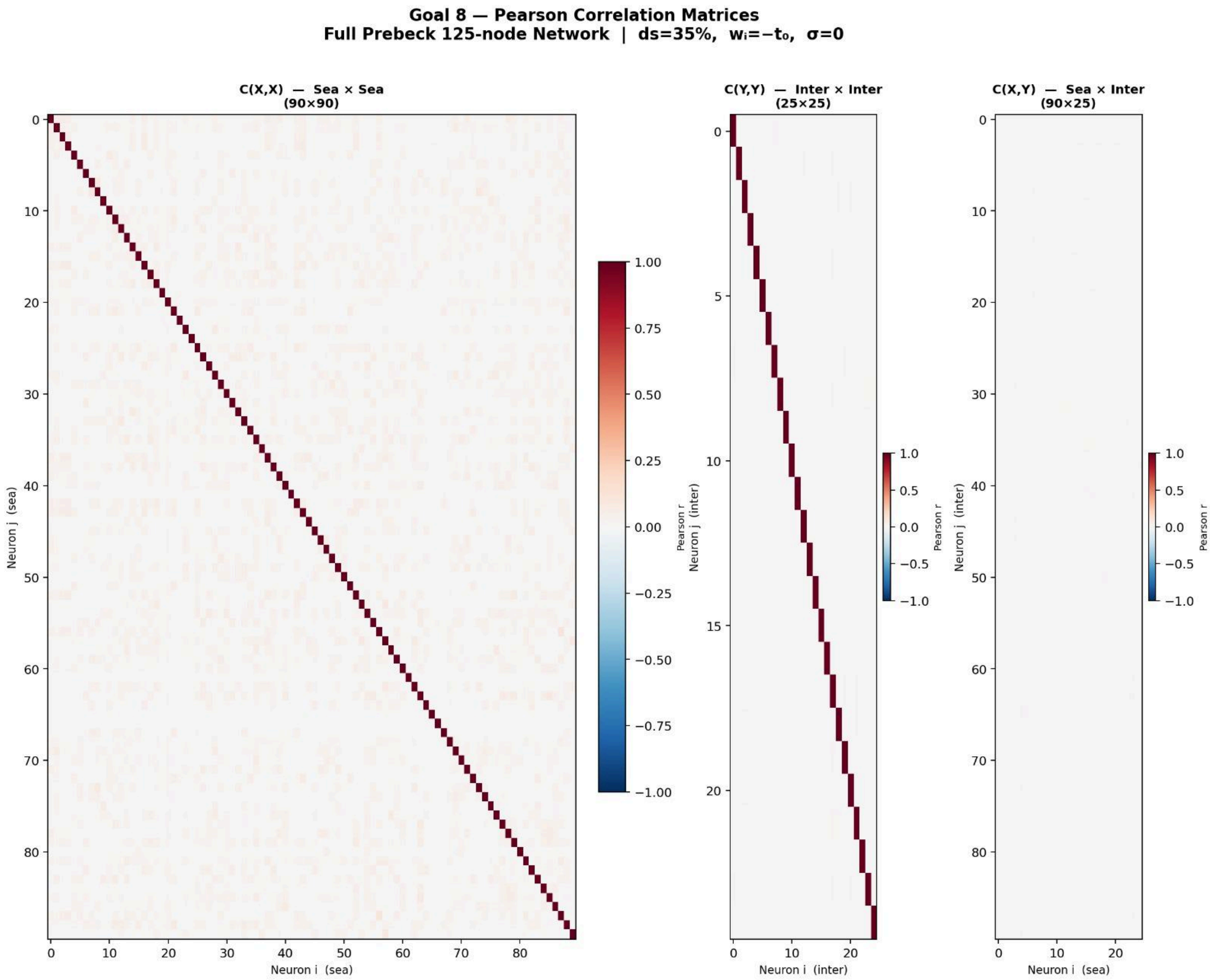
The max. time index is the same for both populations.

We want to compute the matrix C_ij of Pearson correlation coefficients (each coefficient being a correlation over time t) for all neuron pair combinations.

Please compute three such correlation matrices:

C(X,X) of shape 90x90
C(Y,Y) of shape 25x25
C(X,Y) of shape 90x25

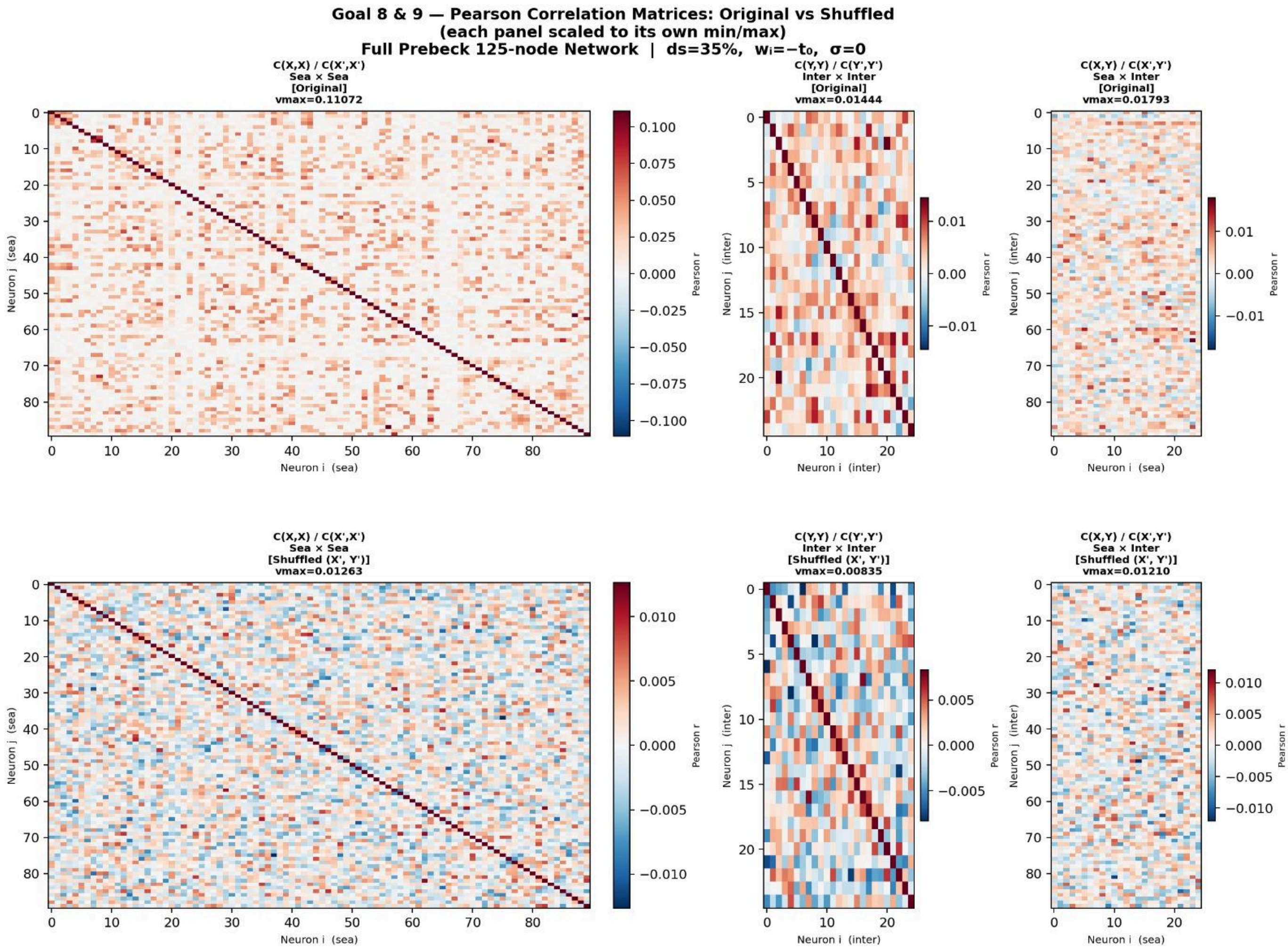
The following function avoids unnecessary multiple computations of the mean and STD of the signals, and takes care of cases where signals are constant:



GOAL 9: Compare Correlations with Shuffled System

It is also interesting to see how the above three correlation matrices look like when the X and Y have the same distribution, but with activations randomly shuffled among time and neuron indices.

Please also show the three correlation matrices C(X',X'), C(Y',Y'), C(X',Y'), where X' is a shuffled version of X.



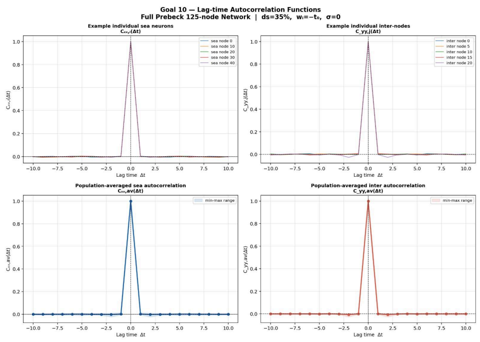
GOAL 10: Lagtime-Dependent Correlation Functions & Pair Probab.

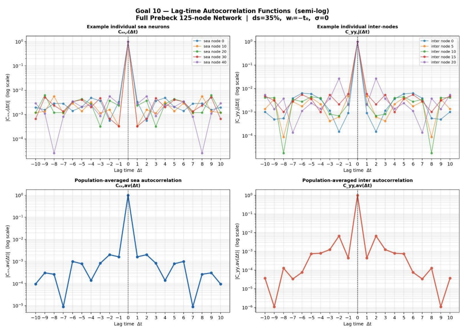
We also need to look at the lag-time dependent auto-correlation functions, like Cxx,i(dt). This would be the Pearson correlation coefficient between X(t,i) and X(t+dt,i).

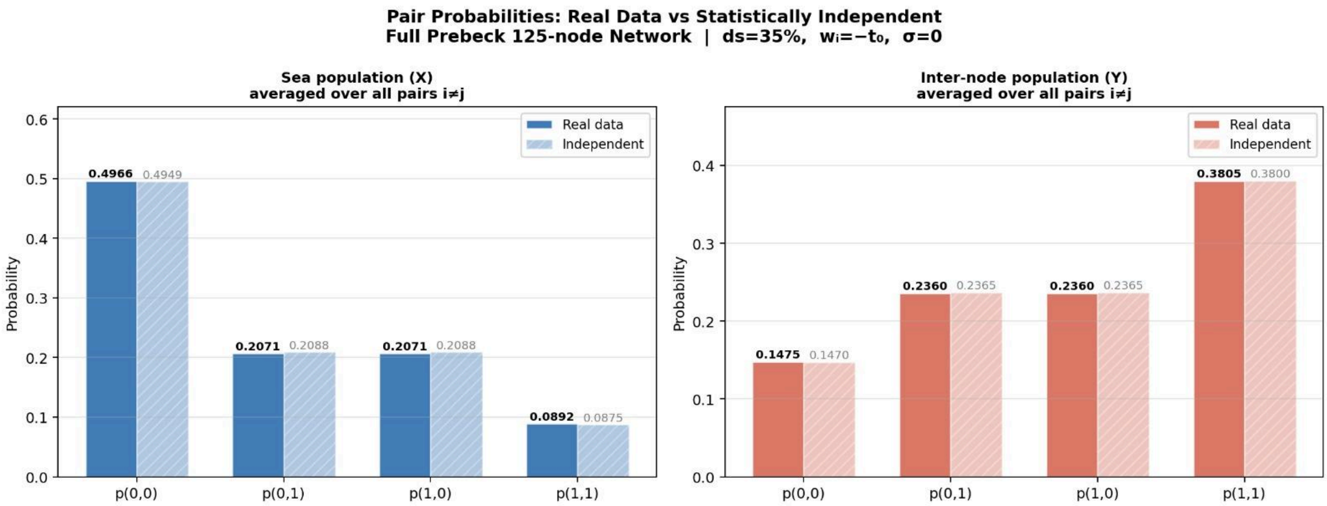
Please show a few examples for Cxx,i(dt) and Cyy,j(dt) for some arbitrary (typical) neurons i. Probably a lag-time range from dt=-10 to dt=+10 is sufficient.

Then calculate the Cxx,i(dt) and Cyy,j(dt) for all i/j, and store the results as two matrices. Then average each matrix over all i/j. This gives two population-averaged functions

Cxx,av(dt) and Cyy,av(dt).

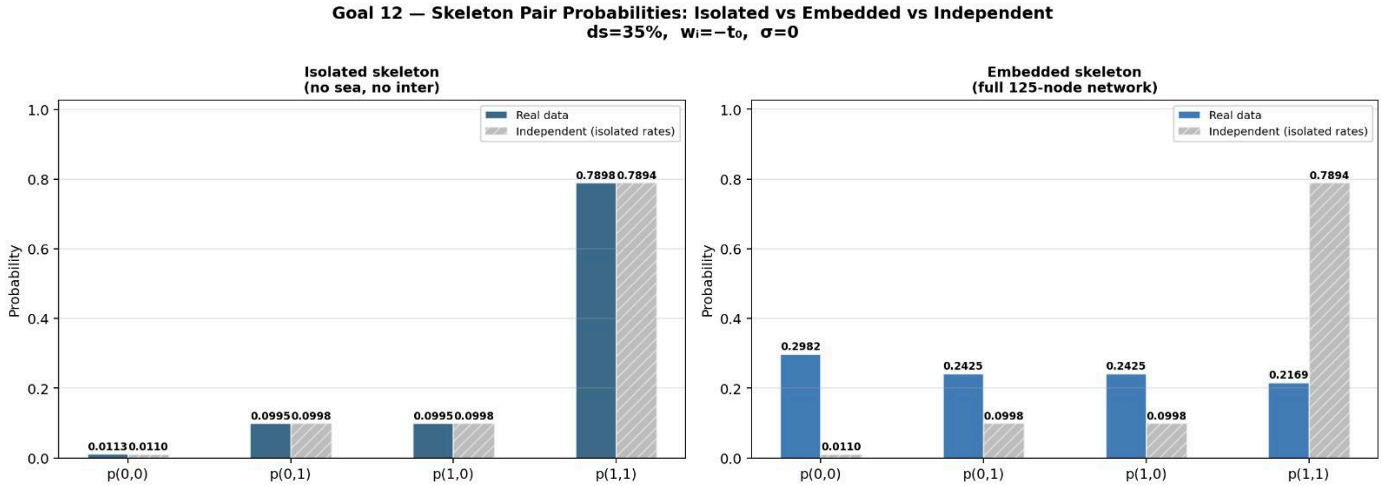






GOAL 11: Pair Probs. in Isolated and Embedded Skeleton

Could you repeat the blue plot when the interneurons are cut away from the skeleton ? I mean just the bare skeleton, without sea and inter. I would like to see if and how the pair probabilities are affected by the inter signals.



GOAL 12: Skeleton-MI as a function of global bias v.

There is only the isolated skeleton. Each of its neurons receives the same constant bias v.

The bias parameter v should be scanned from reasonable negative to positive values.

The plot should contain the horizontal line of the best Prebeck-MI.

Note: This may be too simple. In the Prebeck system, every skeleton neuron is receiving a different signal from the inter-population due to the random coupling matrix, and also these signals are not constant biases.